A LOCAL OBSTRUCTION PROOF OF THE RIEMANN HYPOTHESIS

JOHN SHIELDS

Abstract. We present a local-geometric obstruction proof of the Riemann hypothesis. The two-by-two jet-normalized blocks of the phase kernel of ζ at one or two centers form a finite-dimensional comparison object: a hypothetical off-line zero forces a transverse mode that the actual zeta phase, after a single fixed value-gauge quotient, suppresses at a strictly larger polynomial exponent in the local scale. Two-point and four-point comparisons of this object aggregate over heights via N-point odd-moment cancellation to yield an unconditional zero-free strip around the critical line, hence the Riemann hypothesis. A self-contained final section provides an optional source-energy supply available to the two-point and four-point comparisons if their standalone forms prove insufficient.

Contents

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1. Introduction

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The Riemann hypothesis, originating in Riemann's 1859 memoir [4], asserts that every nontrivial zero of $\zeta(s)$ lies on the critical line $\operatorname{Re} s = \frac{1}{2}$; see [1, 2, 3, 5] for surveys. The proof presented in this paper proceeds by a local-geometric obstruction. For each height γ we work on a smooth window I_T centered at $T = \gamma$ of width comparable to $1/Q(\gamma)$, where Q is a scale parameter slowly growing with $|\gamma|$. On this window we fix a real local phase Φ for ζ and study the phase kernel

$$K_\Phi(x, y) = \frac{\sin\big(\Phi(x) - \Phi(y)\big)}{\pi(x - y)}, \qquad x \neq y,$$

extended by removable singularity at $x = y$. The 2×2 blocks of K_Φ at one or two centers, transformed into the Gram-normalized basis by a fixed elementary matrix P_h, become explicit matrices $J(T)$ and $N_{12}(T_1, T_2)$ whose entries are finite combinations of $q = \Phi'$, q', q'', and $\Delta = \Phi(T_1) - \Phi(T_2)$. All subsequent estimates reduce to finite linear-algebraic manipulations of these matrices.

A hypothetical off-line zero $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$ forces a specific transverse mode in the Gram-normalized jet space — the toy mode — which survives the natural value/gauge quotient. The actual zeta phase, under the same quotient, cannot realize that mode: the transverse channel of the actual whitened block is suppressed at a strictly larger polynomial exponent in Q than the toy lower bound. The clash is two-point at minimum and four-point in the full symmetry quartet $\{\rho, \bar\rho, 1 - \rho, 1 - \bar\rho\}$, propagated across heights by an N-point odd-moment cancellation. Aggregation over the dyadic height ladder gives an unconditional zero-free strip around the critical line; combined with standard low-height verification of RH, this yields the Riemann hypothesis.

The argument is self-contained except for one optional input. If either the two-point or the four-point standalone closure proves insufficient, the source-energy lower bound on the canonical source curvature $q''_{\mathrm{can}} = \partial_t^3 Z$, stated as Hypothesis 12.3 in Section 12, supplies the additional polynomial-in-Q room required.

1.1. Active dependency graph.

The active spine of the argument is the chain

$$\begin{aligned} &\S 2 \longrightarrow \S 3 \longrightarrow \S 4 \\ &\longrightarrow \S 5,\ \S 6,\ \S 7 \\ &\longrightarrow \S 8 \\ &\longrightarrow \S 9 \longrightarrow \S 10 \longrightarrow \S 11. \end{aligned}$$

Section 12 stands aside: it provides the optional source-energy supply consumed by Section 8 when their standalone forms are insufficient. All cross-references to Section 12 from earlier sections are forward references; that section is self-contained.

1.2. Master conditional theorem.

Theorem 1.1 (Master conditional)**.** *Assume the same-point and cross-block jet-limit identities (Lemmas 3.1 and 3.2); the finite-scale whitening loss bound (Lemma 4.8); the toy anomaly visibility lower bound (Theorem 5.14); the two-point or four-point actual-zeta transverse suppression upper bound (Hypothesis 7.1 or Hypothesis 7.3); the corresponding local rigidity theorem (Theorem 8.3 or Theorem 8.6); the branch-exhaustion input (Hypothesis 8.1); the N-point odd-moment cancellation (Lemma 9.1); and the aggregation step (Theorem 10.1). Then every nontrivial zero of ζ satisfies* $\operatorname{Re} s = \frac{1}{2}$.

Gap 1.2. The conditional statement is firm; its inputs are the section-level results listed above. The proof is short — a composition of the section-level results — and is written out in Section 11.

1.3. **Notation and conventions.** We use the symmetric pair convention $T \pm h$ (rather than $T \pm h/2$) throughout. The point-to-jet transform P_h of (2.7) is fixed by this choice. The scale parameter $Q = Q(\gamma)$ is a fixed slowly growing function of the height; all polynomial losses are stated as powers of Q with absolute exponents. For each height parameter T, I_T denotes an interval centered at T of length comparable to $Q(T)^{-1}$: there are fixed absolute constants $0 < c_I \le C_I < \infty$ such that

$$c_I Q(T)^{-1} \le |I_T| \le C_I Q(T)^{-1}.$$

We also require $I_T \subset [T/2, 2T]$; high-height analysis takes T sufficiently large that this inclusion is automatic. The toy polynomial-weight subdomain in Section 5 uses the slow-scale condition $Q(T) = o(q(T))$, where $q = \theta'$; when that subdomain is not being invoked, the relevant lemmas state their window hypotheses explicitly. "Retained packets" refers to a polynomial-weight subfamily of the local packets at height γ on which the chart-regularity and Gram nondegeneracy hypotheses of Definition 2.4 hold.

2. Local kernel and jet normalization

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This section establishes the local geometric machinery used throughout: the Riemann–Siegel phase chart $\Phi_T = \theta$ on the height window I_T of §1.3 (Definition 2.4), the phase kernel K_Φ (Definition 2.7), and the Gram-normalized point-to-jet transform P_h of (2.7).

2.1. The Riemann–Siegel phase chart.

Definition 2.1 (Riemann–Siegel phase)**.** Let $L(z)$ be the unique holomorphic branch of $\log \Gamma z$ on $\Re z > 0$ for which $L(x) = \log \Gamma(x) \in \mathbb{R}$ when $x > 0$. For $t > 0$, define

$$\theta(t) := \Im L\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2}\log \pi, \tag{2.1}$$

and $Z(t) := e^{i\theta(t)}\zeta(1/2 + it)$. The functional equation gives

$$\zeta\left(\tfrac{1}{2} + it\right) = e^{-i\theta(t)}\, Z(t), \qquad Z(t) \in \mathbb{R} \text{ for } t > 0.$$

Remark 2.2 (Branch convention for the Riemann–Siegel phase)**.** The branch L in Definition 2.1 is the holomorphic logarithm of Γ on the half-plane $\Re z > 0$, not the principal logarithm applied pointwise to $\Gamma(z)$. This branch exists because Γ has no zeros or poles on $\Re z > 0$, and the half-plane is simply connected; the normalization on $(0, \infty)$ fixes the additive constant. Any other holomorphic logarithm on a connected subdomain would change θ only by an additive constant $2\pi k$, which leaves $q = \theta'$, the phase differences $\theta(x) - \theta(y)$, and therefore K_θ unchanged. A discontinuous branch, such as a principal-branch representative with jumps, is not part of the definition and cannot be used for derivative identities.

Lemma 2.3 (Differentiated theta asymptotics)**.** *Uniformly for $t \in I_T$,*

$$q(t) := \theta'(t) = \frac{1}{2}\log\frac{t}{2\pi} - \frac{1}{48t^2} + O(t^{-4}), \tag{2.2}$$

$$q'(t) = \theta''(t) = \frac{1}{2t} + O(t^{-3}), \tag{2.3}$$

$$q''(t) = \theta'''(t) = -\frac{1}{2t^2} + O(t^{-4}). \tag{2.4}$$

The constants in the O-terms are absolute.

Proof. Let $z = 1/4 + it/2$. The half-plane $\Re z \ge 1/4$ lies in a fixed Stirling sector. By the differentiated form of Stirling's expansion, after truncating $L(z)$ at z^{-3} the remainder and its t-derivatives up to order three are uniform in this sector, with the corresponding differentiated powers of $|z|$. Taking imaginary parts in (2.1) gives the standard Riemann–Siegel expansion

$$\theta(t) = \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + O(t^{-3}),$$

with a remainder admitting three differentiations in t (needed for $q'' = \theta'''$). Termwise differentiation gives $q(t) = \theta'(t)$; differentiating once and twice more gives $q'(t)$ and $q''(t)$. Since $I_T \subset [T/2, 2T]$, these estimates are uniform on I_T, with absolute constants. $\square$

Definition 2.4 (Local phase chart)**.** A retained high-height local phase chart is the interval I_T together with the real smooth phase

$$\Phi_T(t) := \theta(t) \qquad (t \in I_T).$$

When T is fixed by context, $\Phi := \Phi_T$ and $q := \Phi'$. This is a specialized theta chart, not a general phase-chart axiom. The chart conditions used downstream are:

(P1) $\Phi_T = \theta$ on I_T;

(P2) $q(t) \geq 1$ on I_T, after increasing the global lower-height cutoff if necessary.

If $Q(T) \geq 1$, then (P2) gives the polynomial-in-Q lower bound $q(t) \geq Q(T)^{-c_q}$ with $c_q = 0$. The uniform estimates $q(t) \asymp \log T$, $q'(t) = O(T^{-1})$, and $q''(t) = O(T^{-2})$ on I_T are supplied by Lemma 2.3.

Lemma 2.5 (Phase-derivative lower bound)**.** *Condition* (P2) *of Definition 2.4 holds for all sufficiently large T. More quantitatively, uniformly on I_T,*

$$q(t) \geq \frac{1}{2} \log \frac{T}{4\pi} - C\,T^{-2}$$

for an absolute constant C.

Proof. By Lemma 2.3, $q(t) = \frac{1}{2}\log(t/(2\pi)) + O(t^{-2})$. Since $t \in I_T \subset [T/2, 2T]$, $\log(t/(2\pi)) \geq \log(T/(4\pi))$, and the stated bound follows. The right-hand side is eventually larger than 1. $\square$

Remark 2.6 (No ζ-side regularity is used here)**.** The chart uses ζ only through the identity $\zeta(1/2 + it) = e^{-i\theta(t)} Z(t)$ and through the theta phase. No lower bound on $|Z|$, no zero-free condition in I_T, and no source-energy input is used in Sections 2–3. Such conditions are imposed only in the later sections where actual-zeta transverse suppression or source-energy transfer is consumed.

2.2. **The phase kernel.**

Definition 2.7 (Phase kernel)**.** For a real C^1 phase Φ on an interval I,

$$K_\Phi(x, y) = \begin{cases} \dfrac{\sin\big(\Phi(x) - \Phi(y)\big)}{\pi(x - y)}, & x \neq y, \\[2mm] \dfrac{q(x)}{\pi}, & x = y, \end{cases} \qquad q = \Phi'. \tag{2.5}$$

Lemma 2.8 (Basic kernel properties)**.** *K_Φ is real-valued and symmetric. If $\Phi \in C^1$, the extension across the diagonal is continuous and*

$$K_\Phi(T, T) = \frac{q(T)}{\pi}.$$

If $\Phi \in C^4$, the diagonal first and mixed second derivatives exist and are given by Lemma 2.10.

Proof. For $x \neq y$, symmetry follows from $\sin(\Phi(y) - \Phi(x))/(y - x) = \sin(\Phi(x) - \Phi(y))/(x - y)$. Real-valuedness is immediate. For joint continuity across the diagonal, near any $T \in I$ write

$$\Phi(x) - \Phi(y) = (x - y)A(x, y), \qquad A(x, y) = \int_0^1 q\big(y + \tau(x - y)\big)\, d\tau.$$

The function A is continuous and $A(T, T) = q(T)$. For $x \neq y$,

$$K_\Phi(x, y) = \frac{A(x, y)}{\pi} \frac{\sin\big((x - y)A(x, y)\big)}{(x - y)A(x, y)},$$

with the last factor interpreted by its continuous value 1 at zero. Hence $K_\Phi(x, y) \to q(T)/\pi = K_\Phi(T, T)$ whenever $(x, y) \to (T, T)$. $\square$

Lemma 2.9 (Off-diagonal kernel derivatives)**.** *For $T_1 \neq T_2$ with $s = T_1 - T_2$, $\Delta = \Phi(T_1) - \Phi(T_2)$, $q_j = q(T_j)$, and $\Phi \in C^1$ near T_1, T_2,*

$$K_\Phi(T_1, T_2) = \frac{\sin\Delta}{\pi s}, \qquad K_x(T_1, T_2) = \frac{q_1\, s\cos\Delta - \sin\Delta}{\pi s^2},$$
$$K_y(T_1, T_2) = \frac{\sin\Delta - q_2\, s\cos\Delta}{\pi s^2}, \qquad K_{xy}(T_1, T_2) = \frac{(q_1 + q_2)\, s\cos\Delta + (q_1 q_2 s^2 - 2)\sin\Delta}{\pi s^3}.$$

Proof. Differentiate $K_\Phi(x, y) = \sin(\Phi(x) - \Phi(y))/\pi(x - y)$ by the quotient and chain rules, then evaluate at (T_1, T_2). □

Lemma 2.10 (Diagonal kernel derivatives)**.** *If $\Phi \in C^4$ near T, then*

$$K_\Phi(T, T) = \frac{q(T)}{\pi}, \qquad K_x(T, T) = K_y(T, T) = \frac{q'(T)}{2\pi}, \tag{2.6}$$

$$K_{xy}(T, T) = \frac{q''(T) + 2q(T)^3}{6\pi}.$$

Proof. Set $x = T + u$, $y = T + v$, and write $r = |u| + |v|$; derivatives below are evaluated at T. Factor the phase difference as

$$\Phi(T + u) - \Phi(T + v) = (u - v)A(u, v), \qquad A(u, v) = \int_0^1 q\big(T + v + \tau(u - v)\big)\, d\tau,$$

where the integral form extends continuously to $u = v$. Taylor expansion of the integrand gives

$$A(u, v) = q + \frac{q'}{2}(u + v) + \frac{q''}{6}(u^2 + uv + v^2) + O(r^3).$$

For $u \neq v$, and then by continuous extension to $u = v$, $\Phi(T+u) - \Phi(T+v) = (u-v)A(u, v) = O(r)$ gives

$$\frac{\sin(\Phi(T + u) - \Phi(T + v))}{u - v} = A(u, v) - \frac{(u - v)^2}{6}A(u, v)^3 + O(r^3).$$

Keeping terms through total degree two gives

$$K_\Phi(T + u, T + v) = \frac{1}{\pi}\left[q + \frac{q'}{2}(u + v) + \left(\frac{q''}{6} - \frac{q^3}{6}\right)(u^2 + v^2) + \left(\frac{q''}{6} + \frac{q^3}{3}\right)uv + O(r^3)\right].$$

The coefficients of u, v, and uv are respectively $K_x(T, T)$, $K_y(T, T)$, and $K_{xy}(T, T)$, proving the claim. □

2.3. **Point-to-jet transform.** For symmetric pairs $T \pm h$, define the Gram-normalized coalescing even/odd transform

$$P_h = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1/(2h) & 1/(2h) \end{pmatrix} = \operatorname{diag}(\sqrt{2}, 1/\sqrt{2})\begin{pmatrix} 1/2 & 1/2 \\ -1/(2h) & 1/(2h) \end{pmatrix}. \tag{2.7}$$

Thus P_h is the literal central value/derivative jet transform followed by the fixed diagonal rescaling $\operatorname{diag}(\sqrt{2}, 1/\sqrt{2})$; at the level of Gram blocks this induces the corresponding diagonal congruence. The value/derivative basis produced by P_h is the *Gram-normalized basis.*

With this convention,

$$J(T) = \begin{pmatrix} 2K_\Phi(T, T) & \partial_y K_\Phi(T, T) \\ \partial_x K_\Phi(T, T) & \frac{1}{2}\partial_{xy} K_\Phi(T, T) \end{pmatrix}, \tag{2.8}$$

which is the same-point block computed explicitly in Lemma 3.1.

Remark 2.11 (Pair-separation convention)**.** The pair is $T - h, T + h$, with separation $2h$; the lower row of P_h carries $1/(2h)$. Any formula imported from the alternative $T \pm h/2$ convention or from the literal-jet matrix with lower row $(-1/h, 1/h)$ must be re-conjugated before use.

Remark 2.12 (Convention check)**.** With the literal central value/derivative transform at the pair $T \pm h$, the same-point limit would be

$$\frac{1}{\pi}\begin{pmatrix} q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{6}\big(q''(T)+2q(T)^3\big) \end{pmatrix},$$

related to $J(T)$ by the diagonal congruence $\operatorname{diag}(\sqrt{2}, 1/\sqrt{2})$ of (2.7); the $(1,1)$ entry doubles and the $(2,2)$ entry is halved. All same-point blocks, cross-blocks, finite-scale blocks, whitening matrices, and quotient covectors in this paper are read in the Gram-normalized convention.

3. Jet-limit local blocks

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From the phase kernel K_Φ and the Gram-normalized transform P_h of Section 2, the small-separation limits of K_Φ at one and two centers produce the same-point Gram block $J(T)$ (Lemma 3.1) and the cross-block $N_{12}(T_1, T_2)$ (Lemma 3.2); for the theta phase, $J(T) \succ 0$ at sufficiently large height (Lemma 3.6).

3.1. **Same-point jet limit.**

Lemma 3.1 (Same-point jet limit)**.** *Assume $\Phi \in C^5$ near T. Let*

$$A_h(T) = \begin{pmatrix} K_\Phi(T-h, T-h) & K_\Phi(T-h, T+h) \\ K_\Phi(T+h, T-h) & K_\Phi(T+h, T+h) \end{pmatrix}$$

with row and column order $(T-h, T+h)$. Then

$$P_h\, A_h(T)\, P_h^\top \;=\; J(T) + O(h^2) \qquad (h \to 0),$$

where the $O(h^2)$ is entrywise, with a constant depending only on a fixed neighborhood of T and on a local C^5-bound for Φ. The limiting block is

$$J(T) = \frac{1}{\pi}\begin{pmatrix} 2q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{12}\big(q''(T)+2q(T)^3\big) \end{pmatrix}. \tag{3.1}$$

Proof. Derivatives below are evaluated at T. Taylor expansion gives

$$\Phi(T \pm h) = \Phi(T) \pm hq + \frac{h^2}{2}q' \pm \frac{h^3}{6}q'' + \frac{h^4}{24}q''' + O(h^5),$$

so

$$\Phi(T-h) - \Phi(T+h) = -2hq - \frac{h^3}{3}q'' + O(h^5).$$

Using $\sin z = z - z^3/6 + O(z^5)$,

$$K_\Phi(T-h, T+h) = \frac{q}{\pi} + \frac{h^2}{6\pi}(q'' - 4q^3) + O(h^4). \tag{3.2}$$

Also $K_\Phi(T \pm h, T \pm h) = q(T \pm h)/\pi$, with

$$q(T \pm h) = q \pm hq' + \frac{h^2}{2}q'' \pm \frac{h^3}{6}q''' + O(h^4).$$

The $O(h^5)$ phase remainder and the $O(h^4)$ q-remainder are controlled by a local C^5-bound for Φ; after division by the separation $2h$, the off-diagonal kernel expansion has an $O(h^4)$ remainder, the order needed below since the $(2,2)$ jet entry divides a second difference by h^2. For symmetric M, conjugation by (2.7) gives

$$\begin{aligned}
(P_h M P_h^\top)_{11} &= \tfrac{1}{2}(M_{11} + 2M_{12} + M_{22}), \\
(P_h M P_h^\top)_{12} &= \tfrac{1}{4h}(M_{22} - M_{11}), \\
(P_h M P_h^\top)_{22} &= \tfrac{1}{8h^2}(M_{11} - 2M_{12} + M_{22}).
\end{aligned}$$

Substituting (3.2) and the Taylor series of $q(T \pm h)$: the constant and odd terms cancel in the second difference, while the first difference cancels its constant and even terms. The divisions by h and h^2 therefore leave entrywise $O(h^2)$ remainders:

$$\begin{aligned}(P_h A_h P_h^\top)_{11} &= \frac{2q}{\pi} + O(h^2),\\ (P_h A_h P_h^\top)_{12} &= \frac{q'}{2\pi} + O(h^2),\\ (P_h A_h P_h^\top)_{22} &= \frac{q'' + 2q^3}{12\pi} + O(h^2),\end{aligned}$$

which is (3.1). □

3.2. **Cross-block jet limit.**

Lemma 3.2 (Cross-block jet limit)**.** *Assume $\Phi \in C^5$ in neighborhoods of T_1 and T_2, let $T_1 \neq T_2$, set*

$$s = T_1 - T_2, \qquad \Delta = \Phi(T_1) - \Phi(T_2), \qquad q_j = q(T_j),$$

and assume $0 < h < |s|/3$, with h also sufficiently small that the four sampled points lie in the chosen neighborhoods of T_1 and T_2. Define

$$C_h(T_1, T_2) = \begin{pmatrix} K_\Phi(T_1 - h, T_2 - h) & K_\Phi(T_1 - h, T_2 + h) \\ K_\Phi(T_1 + h, T_2 - h) & K_\Phi(T_1 + h, T_2 + h)\end{pmatrix}$$

with row order $(T_1 - h, T_1 + h)$ and column order $(T_2 - h, T_2 + h)$. Then

$$P_h\, C_h(T_1, T_2)\, P_h^\top \;=\; \frac{1}{\pi} N_{12}(T_1, T_2) + O(h^2) \qquad (h \to 0),$$

where the $O(h^2)$ is entrywise for fixed $T_1 \neq T_2$; the implied constant may depend on a finite power of $|s|^{-1}$ and on local C^5-bounds for Φ, but not on h. Here

$$N_{12} = \begin{pmatrix} \dfrac{2\sin\Delta}{s} & \dfrac{\sin\Delta - q_2 s\cos\Delta}{s^2} \\ \dfrac{q_1 s\cos\Delta - \sin\Delta}{s^2} & \dfrac{(q_1 + q_2)s\cos\Delta + (q_1 q_2 s^2 - 2)\sin\Delta}{2s^3}\end{pmatrix}. \tag{3.3}$$

Proof. The kernel is smooth in a neighborhood of (T_1, T_2), and the hypothesis $h < |s|/3$ gives $|s + u_1 - u_2| \geq |s|/3$ for $|u_1|, |u_2| \leq h$, bounding the Taylor remainder on the sampled square. Write $K_{ab} = \partial_x^a \partial_y^b K_\Phi(T_1, T_2)$. Taylor expansion to total order four gives, uniformly for $u_1, u_2 \in \{-h, +h\}$,

$$K_\Phi(T_1 + u_1, T_2 + u_2) = \sum_{a+b\leq 4} \frac{K_{ab}}{a!\,b!}\, u_1^a\, u_2^b + O_s(h^5).$$

The subscript indicates that the constant may depend on a finite power of $|s|^{-1}$ and on local C^5-bounds for Φ in the chosen off-diagonal neighborhoods. For a general 2×2 matrix M,

$$\begin{aligned}(P_h M P_h^\top)_{11} &= \tfrac{1}{2}(M_{11} + M_{12} + M_{21} + M_{22}),\\ (P_h M P_h^\top)_{12} &= \tfrac{1}{4h}(-M_{11} + M_{12} - M_{21} + M_{22}),\\ (P_h M P_h^\top)_{21} &= \tfrac{1}{4h}(-M_{11} - M_{12} + M_{21} + M_{22}),\\ (P_h M P_h^\top)_{22} &= \tfrac{1}{8h^2}(M_{11} - M_{12} - M_{21} + M_{22}).\end{aligned}$$

Equivalently, these four expressions apply the sign weights 1, σ_2, σ_1, $\sigma_1\sigma_2$ respectively to the four samples with $u_i = \sigma_i h$. Parity selects the required leading terms: the next surviving polynomial monomials in the $(1,1)$ entry have total degree at least two, contributing $O(h^2)$; the $(1,2)$ and $(2,1)$ numerators select monomials with odd degree in one variable, so after division by h the surviving polynomial corrections are $O(h^2)$; the $(2,2)$ numerator selects monomials odd in both variables, and division by h^2 again leaves $O(h^2)$. The Taylor remainder $O_s(h^5)$ per sample contributes $O_s(h^5)$ to the $(1,1)$ entry (no division), $O_s(h^4)$ to the off-diagonal entries (division by $4h$), and $O_s(h^3)$ to the $(2,2)$ entry (division by $8h^2$). Hence

$$(P_h C_h P_h^\top)_{11} = 2K_{00} + O(h^2), \qquad (P_h C_h P_h^\top)_{12} = K_{01} + O(h^2),$$

$$(P_h C_h P_h^\top)_{21} = K_{10} + O(h^2), \qquad (P_h C_h P_h^\top)_{22} = \tfrac{1}{2}K_{11} + O(h^2).$$

Substituting the off-diagonal derivative formulas of Lemma 2.9 gives (3.3). ☐

Remark 3.3 (Fixed-separation scope)**.** Lemma 3.2 is an off-diagonal jet limit at fixed separation. It is not a collision expansion for $T_1 \to T_2$; the constants are not asserted to be uniform as $s = T_1 - T_2 \to 0$. Any two-center blow-up or near-coalescence analysis must be derived from a separate scaling argument, rather than by citing this lemma with $|s|$ comparable to h.

3.3. **Same-point Gram positivity.**

Definition 3.4 (Same-point Gram determinant)**.**

$$D_J(T) \;:=\; 4\,q(T)^4 + 2\,q(T)\,q''(T) - 3\,q'(T)^2. \tag{3.4}$$

Lemma 3.5 (Algebraic same-point Gram criterion)**.** *For $J(T)$ of* (3.1)*,*

$$\operatorname{Tr} J(T) = \frac{2q(T)^3 + 24\,q(T) + q''(T)}{12\pi}, \qquad \det J(T) = \frac{D_J(T)}{12\pi^2}. \tag{3.5}$$

Moreover,

$$J(T) \succ 0 \quad \Longleftrightarrow \quad q(T) > 0 \quad \text{and} \quad D_J(T) > 0.$$

Proof. The trace and determinant follow by direct computation from (3.1). Since $J(T)$ is real symmetric, Sylvester's criterion gives positive definiteness exactly when its leading principal minor $2q(T)/\pi$ and its determinant are positive. ☐

Lemma 3.6 (Same-point Gram positivity for the theta phase)**.** *For $\Phi_T = \theta$, and for all sufficiently large retained heights, $J(t)$ is positive definite uniformly for $t \in I_T$. More precisely,*

$$\lambda_{\min}(J(t)) = \frac{2q(t)}{\pi}\,(1 + o_T(1)) = \frac{1}{\pi}\,\log\frac{t}{2\pi}\,(1 + o_T(1)) \qquad (t \in I_T), \tag{3.6}$$

where the error is uniform on I_T. In particular this holds at the center $t = T$. After increasing the lower-height cutoff if necessary, $\lambda_{\min}(J(t)) \ge 1$ uniformly on I_T; hence the polynomial-in-Q lower bound $\lambda_{\min}(J(t)) \ge Q(T)^{-C_J}$ holds with $C_J = 0$ whenever $Q(T) \ge 1$. The cutoff is qualitative; no explicit numerical lower height is claimed here.

Proof. Let $t \in I_T$. By Lemma 2.3, uniformly on I_T,

$$q(t) = \tfrac{1}{2}\log(t/(2\pi)) + O(t^{-2}), \qquad q'(t) = O(t^{-1}), \qquad q''(t) = O(t^{-2}).$$

Therefore, uniformly on I_T,

$$D_J(t) = 4\,q(t)^4 + O\big(q(t)\,t^{-2}\big) + O(t^{-2}) = 4\,q(t)^4\,(1 + o_T(1)) > 0$$

for sufficiently large T, and $q(t) > 0$. By Lemma 3.5, $J(t) \succ 0$.

Let $0 < \lambda_1(t) \le \lambda_2(t)$ be the eigenvalues of $J(t)$. Since $\lambda_2(t) \le \operatorname{Tr} J(t)$,

$$\lambda_1(t) = \frac{\det J(t)}{\lambda_2(t)} \ge \frac{\det J(t)}{\operatorname{Tr} J(t)} = \frac{D_J(t)}{\pi(2q(t)^3 + 24\,q(t) + q''(t))} = \frac{2q(t)}{\pi}\,(1 + o_T(1)),$$

uniformly for $t \in I_T$. The Rayleigh quotient at e_1 gives $\lambda_1(t) \le J_{11}(t) = 2q(t)/\pi$, proving (3.6). ☐

Remark 3.7 (Scope of the positivity lemma)**.** Lemma 3.6 is theta-specific. For an arbitrary real phase, the safe statement is only the algebraic criterion $q > 0$ and $D_J > 0$. This lemma does not prove any lower bound for unrelated anisotropy functionals such as $q'' - 6q + 2q^3$, nor does it supply source-energy input of the kind consumed in Section 12.

4. Finite-scale local blocks and whitening

paper: [**LIVE**] v1: [**migrated**] referee: [**in-progress**] sympy: [**verified**] rh/sympy/sec-finite-scale-local-blocks-and-whitening
sim: [**verified**] rh/simulation/sec-finite-scale-local-blocks-and-whitening.py lean: [**n/a**]

Around a midpoint m at finite separation s, the same-point and cross-block jets of Section 3 give two same-point Gram blocks $G_{m,\pm}(s)$ and a mixed block $N_m(s)$. This section records the finite real-axis block package and the whitening-factor loss used by later comparison arguments.

Definition 4.1 (Finite packet domain)**.** For a retained theta chart I_T, set $t_\pm = m \pm s/2$ and

$$\mathcal{D}_T := \big\{(m,s) \in \mathbb{R}^2 : t_- \in I_T,\ t_+ \in I_T\big\}.$$

Write $\mathcal{D}_T^\times := \{(m,s) \in \mathcal{D}_T : s \neq 0\}$. The orientation convention is that rows correspond to the center t_- and columns to the center t_+. In applications $s > 0$, but allowing signed s makes the removable-singularity statement at $s = 0$ precise. Uniform estimates below are taken on $\mathcal{D}_T$, after the global lower-height cutoff has been enlarged if necessary.

4.1. Finite same-point and mixed blocks.

For $(m,s) \in \mathcal{D}_T$, write

$$t_\pm = m \pm s/2, \qquad q_\pm = q(t_\pm), \qquad \Delta_m(s) = \Phi(t_-) - \Phi(t_+).$$

Definition 4.2 (Finite same-point and mixed blocks)**.** For $(m,s) \in \mathcal{D}_T$, the finite same-point blocks are

$$G_{m,\pm}(s) = \frac{1}{\pi}\begin{pmatrix} 2q(t_\pm) & \frac{1}{2}q'(t_\pm) \\ \frac{1}{2}q'(t_\pm) & \frac{1}{12}\big(q''(t_\pm) + 2q(t_\pm)^3\big)\end{pmatrix}. \tag{4.1}$$

For $s \neq 0$, the finite mixed block is

$$N_m(s) = \frac{1}{\pi}\begin{pmatrix} -\dfrac{2\sin\Delta_m(s)}{s} & \dfrac{\sin\Delta_m(s) + q_+ s\cos\Delta_m(s)}{s^2} \\ -\dfrac{\sin\Delta_m(s) + q_- s\cos\Delta_m(s)}{s^2} & \dfrac{(q_- + q_+)s\cos\Delta_m(s) + (2 - q_-q_+s^2)\sin\Delta_m(s)}{2s^3}\end{pmatrix}. \tag{4.2}$$

Lemma 4.3 (Exact finite-block interpretation)**.** *For $(m,s) \in \mathcal{D}_T$,*

$$G_{m,\pm}(s) = J(t_\pm).$$

For $(m,s) \in \mathcal{D}_T^\times$, moreover,

$$N_m(s) = \begin{pmatrix} 2K_\Phi(t_-,t_+) & \partial_y K_\Phi(t_-,t_+) \\ \partial_x K_\Phi(t_-,t_+) & \frac{1}{2}\partial_{xy}K_\Phi(t_-,t_+)\end{pmatrix}. \tag{4.3}$$

Equivalently, $N_m(s) = \pi^{-1}N_{12}$ for the matrix N_{12} of (3.3) taken at $(T_1,T_2) = (t_-,t_+)$, with $s_{12} = T_1 - T_2 = -s$ absorbed into the displayed signs in (4.2).

Proof. The identity $G_{m,\pm}(s) = J(t_\pm)$ is immediate by comparing (4.1) with (3.1). For the mixed block, apply the off-diagonal kernel-derivative formulas of Lemma 2.9 at $(T_1,T_2) = (t_-,t_+)$; there $s_{12} = t_- - t_+ = -s$, $q_1 = q_-$, $q_2 = q_+$, and $\Delta = \Delta_m(s)$. Substitution gives the four entries of (4.3), which match (4.2) entry by entry. □

Lemma 4.4 (Removable singularity of $N_m(s)$)**.** *Let $m \in I_T$. If $\Phi \in C^3$ on a neighborhood of m, then, as $s \to 0$ with $(m,s) \in \mathcal{D}_T^\times$, the entries of $N_m(s)$ of (4.2) extend continuously to $s = 0$ with*

$$N_m(0) = J(m) = \frac{1}{\pi}\begin{pmatrix} 2q(m) & \frac{1}{2}q'(m) \\ \frac{1}{2}q'(m) & \frac{1}{12}\big(q''(m) + 2q(m)^3\big)\end{pmatrix}. \tag{4.4}$$

Proof. Write $q = q(m)$, $a = q'(m)$, $b = q''(m)$. Taylor expansion about m, using the symmetric points $m \pm s/2$, gives

$$\Delta_m(s) = \Phi(m - s/2) - \Phi(m + s/2) = -qs - \frac{b}{24}s^3 + o(s^3),$$

$$q_+ = q + \frac{a}{2}s + \frac{b}{8}s^2 + o(s^2), \qquad q_- = q - \frac{a}{2}s + \frac{b}{8}s^2 + o(s^2).$$

Hence

$$\sin\Delta_m(s) = -qs + \left(\frac{q^3}{6} - \frac{b}{24}\right)s^3 + o(s^3), \qquad \cos\Delta_m(s) = 1 - \frac{q^2}{2}s^2 + o(s^2).$$

Substituting into (4.2) entry by entry,

$$\begin{aligned} -\frac{2\sin\Delta_m(s)}{s} &= 2q + o(1), \\ \frac{\sin\Delta_m(s) + q_+ s\cos\Delta_m(s)}{s^2} &= \frac{a}{2} + o(1), \\ -\frac{\sin\Delta_m(s) + q_- s\cos\Delta_m(s)}{s^2} &= \frac{a}{2} + o(1), \\ \frac{(q_- + q_+)s\cos\Delta_m(s) + (2 - q_-q_+s^2)\sin\Delta_m(s)}{2s^3} &= \frac{b}{12} + \frac{q^3}{6} + o(1). \end{aligned}$$

The common factor $1/\pi$ in (4.2) gives (4.4). □

Remark 4.5 (Coalescence is not the cross-block jet limit)**.** Lemma 3.2 is a fixed-center-separation limit in the sampling parameter h; its constants are not uniform as $s \to 0$ (Remark 3.3). Lemma 4.4 is a separate coalescence statement in the center-separation parameter s, proved by direct Taylor expansion above. From this point on, $N_m(0)$ denotes the continuous extension supplied by Lemma 4.4.

4.2. **Whitened finite block.**

Definition 4.6 (Whitened finite block)**.** For $(m, s) \in \mathcal{D}_T$, with $N_m(0)$ understood by continuous extension, whenever $G_{m,-}(s)$ and $G_{m,+}(s)$ are positive definite, let $G_{m,\pm}(s)^{-1/2}$ denote the positive definite inverse square root and set

$$\widehat{\Omega}_\zeta(s; m) := G_{m,-}(s)^{-1/2}\, N_m(s)\, G_{m,+}(s)^{-1/2}. \tag{4.5}$$

Corollary 4.7 (Coalescence: $\widehat{\Omega}_\zeta(0; m) = I_2$)**.** *For every $m \in I_T$ at sufficiently large height, $\widehat{\Omega}_\zeta(s; m)$ extends continuously to $s = 0$, along $(m, s) \in \mathcal{D}_T$, with $\widehat{\Omega}_\zeta(0; m) = I_2$.*

Proof. By Lemma 3.6, $J(m) \succ 0$ at every midpoint $m \in I_T$ at sufficiently large height. By continuity, $G_{m,\pm}(s) = J(m \pm s/2)$ remain positive definite for all sufficiently small $|s|$ with $(m, s) \in \mathcal{D}_T$, and the map $G \mapsto G^{-1/2}$ is continuous on the positive definite cone. Lemma 4.4 gives $N_m(0) = J(m)$, while $G_{m,\pm}(0) = J(m)$ by Lemma 4.3. Therefore $\widehat{\Omega}_\zeta(0; m) = J(m)^{-1/2}\, J(m)\, J(m)^{-1/2} = I_2$. □

4.3. **Whitening loss.**

Lemma 4.8 (Operator-norm bound for $G_{m,\pm}^{-1/2}$)**.** *For $\Phi_T = \theta$ on I_T, after enlarging the global lower-height cutoff if necessary, $G_{m,\pm}(s)$ are positive definite for every $(m, s) \in \mathcal{D}_T$, and there is a quantity $\varepsilon_T \to 0$, independent of $(m, s) \in \mathcal{D}_T$, such that*

$$\left\|G_{m,\pm}(s)^{-1/2}\right\|_{\mathrm{op}} \;\le\; \sqrt{\frac{\pi}{2\, q(t_\pm)}}\,(1 + \varepsilon_T) \qquad (T \to \infty). \tag{4.6}$$

After possibly enlarging the same lower-height cutoff,

$$\left\|G_{m,\pm}(s)^{-1/2}\right\|_{\mathrm{op}} \;\le\; Q(T)^{C_W} \tag{4.7}$$

uniformly on $\mathcal{D}_T$, with absolute exponent $C_W = 0$ whenever $Q(T) \ge 1$.

Proof. By Lemma 4.3, $G_{m,\pm}(s) = J(t_\pm)$; since $(m, s) \in \mathcal{D}_T$ implies $t_\pm \in I_T \subset [T/2, 2T]$, the theta-derivative estimates of Lemma 2.3 are uniform at $t = t_\pm$. The proof of Lemma 3.6 then applies with T replaced by an arbitrary $t \in I_T$, giving, uniformly on I_T,

$$\lambda_{\min}(J(t)) = \frac{2q(t)}{\pi}\,(1 + o(1)).$$

Equivalently, for some $\delta_T \to 0$ independent of $t \in I_T$,

$$\lambda_{\min}(J(t)) \geq \frac{2q(t)}{\pi}(1-\delta_T).$$

For large T, $1-\delta_T > 0$; the finite same-point blocks are positive definite, and

$$\left\|G_{m,\pm}(s)^{-1/2}\right\|_{\mathrm{op}} = \lambda_{\min}(G_{m,\pm}(s))^{-1/2} \leq \sqrt{\frac{\pi}{2\,q(t_\pm)}}\,(1-\delta_T)^{-1/2}.$$

Absorbing $(1-\delta_T)^{-1/2}$ into $1+\varepsilon_T$ gives (4.6).

For the polynomial bound, the uniform asymptotic above together with Lemma 2.5 gives $\lambda_{\min}(J(t)) \to \infty$ uniformly for $t \in I_T$; after enlarging the lower-height cutoff, $\lambda_{\min}(J(t)) \geq 1$ uniformly on I_T. Hence $\|G_{m,\pm}(s)^{-1/2}\|_{\mathrm{op}} \leq 1 \leq Q(T)^0$ whenever $Q(T) \geq 1$, which is (4.7) with $C_W = 0$. □

Remark 4.9 (Scope of Lemma 4.8)**.** Lemma 4.8 controls only the real-axis same-point whitening factors attached to the theta phase. It does not prove actual-zeta transverse suppression, does not supply source-energy input, does not give a uniform bound on $\widehat{\Omega}_\zeta(s;m)$ itself beyond the whitening-factor loss, and does not give complex-disk or holomorphic whitening estimates. Any stronger whitening statement needed for a later complex-analytic argument must be introduced as a separate input or proved in a separate section.

4.4. **Two-sided whitening ledger.** The one-sided estimate of Lemma 4.8 prevents same-point whitening from enlarging upper bounds. Lower bounds require a separate ledger, because left and right multiplication by $G_{m,\pm}^{-1/2}$ can attenuate a nonzero matrix. Define the whitening map on 2×2 real matrices by

$$\mathsf{W}_{m,s}(X) := G_{m,-}(s)^{-1/2}\,X\,G_{m,+}(s)^{-1/2}. \tag{4.8}$$

Lemma 4.10 (Spectral size of the finite same-point blocks)**.** *For $\Phi_T = \theta$, uniformly for $(m,s) \in \mathcal{D}_T$,*

$$\lambda_{\min}\big(G_{m,\pm}(s)\big) = \frac{2q(t_\pm)}{\pi}(1+o_T(1)), \tag{4.9}$$

$$\lambda_{\max}\big(G_{m,\pm}(s)\big) = \frac{q(t_\pm)^3}{6\pi}(1+o_T(1)). \tag{4.10}$$

Consequently, after increasing the global lower-height cutoff, there are absolute constants $c, C > 0$ such that, uniformly on $\mathcal{D}_T$,

$$c\,q(t_\pm) \leq \lambda_{\min}\big(G_{m,\pm}(s)\big) \leq C\,q(t_\pm), \qquad c\,q(t_\pm)^3 \leq \lambda_{\max}\big(G_{m,\pm}(s)\big) \leq C\,q(t_\pm)^3. \tag{4.11}$$

Proof. By Lemma 4.3, $G_{m,\pm}(s) = J(t_\pm)$. Since $(m,s) \in \mathcal{D}_T$, both $t_\pm \in I_T$, and the theta derivative asymptotics of Lemma 2.3 are uniform at these two points. The proof of Lemma 3.6 therefore applies uniformly with T replaced by any $t \in I_T$, giving (4.9).

For the largest eigenvalue, use

$$\lambda_{\max}(J(t)) = \mathrm{Tr}\,J(t) - \lambda_{\min}(J(t)).$$

By (3.5), uniformly for $t \in I_T$,

$$\mathrm{Tr}\,J(t) = \frac{2q(t)^3 + 24q(t) + q''(t)}{12\pi} = \frac{q(t)^3}{6\pi}(1+o_T(1)),$$

while $\lambda_{\min}(J(t)) = O(q(t))$. Since $q(t) \to \infty$ uniformly on I_T, the trace term of size $q(t)^3$ dominates, which gives (4.10). The two-sided comparability bounds (4.11) follow by enlarging the lower-height cutoff once more. □

Lemma 4.11 (Two-sided whitening ledger)**.** *For $\Phi_T = \theta$, after increasing the global lower-height cutoff, there are absolute constants $c, C > 0$ such that for every $(m,s) \in \mathcal{D}_T$ and every real 2×2 matrix X,*

$$c\,\big(q(t_-)q(t_+)\big)^{-3/2}\,\|X\|_{\mathrm{op}} \leq \|\mathsf{W}_{m,s}(X)\|_{\mathrm{op}} \leq C\,\big(q(t_-)q(t_+)\big)^{-1/2}\,\|X\|_{\mathrm{op}}. \tag{4.12}$$

In particular, the lower-bound attenuation of two-sided whitening is at worst

$$\big(q(t_-)q(t_+)\big)^{-3/2} \asymp (\log T)^{-3} \tag{4.13}$$

uniformly on $\mathcal{D}_T$.

Proof. Let

$$A = G_{m,-}(s)^{-1/2}, \qquad B = G_{m,+}(s)^{-1/2}.$$

The upper bound follows from submultiplicativity:

$$\|AXB\|_{\mathrm{op}} \le \|A\|_{\mathrm{op}}\,\|X\|_{\mathrm{op}}\,\|B\|_{\mathrm{op}} = \lambda_{\min}(G_{m,-})^{-1/2}\lambda_{\min}(G_{m,+})^{-1/2}\,\|X\|_{\mathrm{op}}.$$

Using the lower-eigenvalue estimate in Lemma 4.10 gives the right-hand inequality in (4.12).

For the lower bound, write

$$X = A^{-1}(AXB)B^{-1}.$$

Therefore

$$\|X\|_{\mathrm{op}} \le \|A^{-1}\|_{\mathrm{op}}\,\|AXB\|_{\mathrm{op}}\,\|B^{-1}\|_{\mathrm{op}} = \lambda_{\max}(G_{m,-})^{1/2}\lambda_{\max}(G_{m,+})^{1/2}\,\|AXB\|_{\mathrm{op}}.$$

The upper-eigenvalue estimate in Lemma 4.10 gives

$$\|AXB\|_{\mathrm{op}} \ge c\,\big(q(t_-)q(t_+)\big)^{-3/2}\,\|X\|_{\mathrm{op}},$$

which is the left-hand inequality in (4.12). Finally, Lemma 2.3 gives $q(t_\pm) \asymp \log T$ uniformly on I_T, and hence (4.13). □

Corollary 4.12 (Polynomial ledger form)**.** *Assume the packet scale has an upper phase-slope ledger*

$$q(t) \le Q(T)^{C_q^+} \qquad (t \in I_T) \tag{4.14}$$

for some absolute exponent C_q^+*. Then for all retained* $(m,s) \in \mathcal{D}_T$ *and all real* 2×2 *matrices* X,

$$\|\mathsf{W}_{m,s}(X)\|_{\mathrm{op}} \ge c\,Q(T)^{-3C_q^+}\,\|X\|_{\mathrm{op}}, \tag{4.15}$$

with an absolute constant $c > 0$*. Thus any pre-whitening lower bound* $\|X\|_{\mathrm{op}} \ge Q(T)^{-A}$ *becomes the post-whitening lower bound*

$$\|\mathsf{W}_{m,s}(X)\|_{\mathrm{op}} \ge c\,Q(T)^{-(A+3C_q^+)}. \tag{4.16}$$

Proof. Combine the lower bound in Lemma 4.11 with (4.14) at $t_\pm$:

$$\big(q(t_-)q(t_+)\big)^{-3/2} \ge Q(T)^{-3C_q^+}.$$

This gives (4.15); applying it to a matrix satisfying $\|X\|_{\mathrm{op}} \ge Q(T)^{-A}$ gives (4.16). □

Remark 4.13 (When the ledger is consumed)**.** Lemma 4.11 is needed only when a toy, source-side, or finite-algebraic lower bound is first proved in unwhitened coordinates and then transported through $\mathsf{W}_{m,s}$. If a visibility theorem is stated directly for the already whitened block $\widehat{\Omega}_\zeta$, then no additional whitening loss is charged. The same caveat applies to any fixed finite-dimensional norm other than $\|\cdot\|_{\mathrm{op}}$: because all norms on $M_2(\mathbb{R})$ are equivalent, the constants in the ledger change, but the powers of $q(t_\pm)$, $\log T$, and $Q(T)$ do not.

5. Toy anomaly and transverse obstruction

paper: **[LIVE]** v1: **[migrated]** referee: **[in-progress]** sympy: **[verified]** `rh/sympy/sec-toy-anomaly-and-transverse-obstruction.`
sim: **[verified]** `rh/simulation/sec-toy-anomaly-and-transverse-obstruction.py` lean: **[not-started]**

The toy phase Φ_{toy} defined below is a real-analytic visibility model carrying an explicit even-u^2 deformation of the linear baseline, with $u = \beta - \frac{1}{2}$ the off-line displacement parameter. Applying the finite-block package of Section 4 to Φ_{toy} produces a toy whitened block $\widehat{\Omega}_{\mathrm{toy}}(s; m, u, q_0)$. The off-diagonal trace $\mathcal{A}_{\mathrm{toy}}$ of this block is identically zero on the on-line baseline and has nonzero leading-u^2 coefficient $F_{\mathrm{toy}}(d, q_0)$, where $d = q_0\, s$ is the dimensionless local separation. As $q_0 \to \infty$, this coefficient converges uniformly on compact d-intervals to a q_0-independent closed-form limit $F_\infty(d)$. The gauge-fixing projection Π_{trans} of Section 6 is a separate step; this section proves the scalar visibility input that such a projection must preserve.

5.1. **Off-line toy phase.**

Definition 5.1 (Off-line toy phase)**.** Fix a height m_0, a local frequency $q_0 > 0$, and an off-line displacement $u \in \mathbb{R}$. The off-line toy phase is the real-analytic phase

$$\Phi_{\mathrm{toy}}(t; u, m_0, q_0) \;:=\; q_0\,(t - m_0) \;+\; u^2\,\big(1 - \cos(q_0\,(t - m_0))\big), \qquad t \in \mathbb{R}. \tag{5.1}$$

The displacement is identified with the off-line real part by $u = \beta - \frac{1}{2}$. By construction Φ_{toy} depends on u only through u^2 and is C^∞ on $\mathbb{R}$ for every (u, q_0).

Remark 5.2 (Toy frequency convention)**.** The parameter q_0 is part of the toy data. When it matters, we write

$$G_{\mathrm{toy},\pm}(s; m, u, q_0), \qquad N_{\mathrm{toy}}(s; m, u, q_0), \qquad \widehat{\Omega}_{\mathrm{toy}}(s; m, u, q_0).$$

In Section 5.4, where the toy is placed on a retained theta packet, the toy frequency is specialized to the local theta slope $q_0 = q(m)$. This convention is essential: the toy visibility theorem is uniform in an abstract high-frequency parameter q_0, while the polynomial-weight subdomain chooses that parameter from the actual retained packet.

Lemma 5.3 (Toy phase derivatives)**.** *With $\tau := t - m_0$, the derivatives of Φ_{toy} are*

$$q_{\mathrm{toy}}(\tau) = q_0 + u^2\,q_0 \sin(q_0\tau), \quad q'_{\mathrm{toy}}(\tau) = u^2\,q_0^2 \cos(q_0\tau), \quad q''_{\mathrm{toy}}(\tau) = -u^2\,q_0^3 \sin(q_0\tau). \tag{5.2}$$

At $\tau = 0$, $q_{\mathrm{toy}}(0) = q_0$, $q'_{\mathrm{toy}}(0) = u^2 q_0^2$, $q''_{\mathrm{toy}}(0) = 0$. The toy phase is not the retained theta chart of Definition 2.4; it is used only as a model phase. Its positivity analogue is

$$q_{\mathrm{toy}}(\tau) \geq q_0(1 - u^2),$$

so the toy same-point blocks are safely in the positive-frequency regime whenever $|u| < 1$ and $q_0(1 - u^2) > 0$.

Proof. Direct differentiation of (5.1). The lower bound $q_{\mathrm{toy}} \geq q_0(1 - u^2)$ follows from $|\sin| \leq 1$. ☐

Lemma 5.4 (Symmetric-centre splitting at scale s)**.** *With $t_\pm = m_0 \pm s/2$ and $d := q_0\,s$, the toy phase derivatives at the symmetric centres satisfy*

$$q_\pm := q_{\mathrm{toy}}(\pm s/2) \;=\; q_0 \;\pm\; u^2\,q_0 \sin(d/2), \qquad q'_\pm \;=\; u^2\,q_0^2 \cos(d/2), \tag{5.3}$$

$$q''_\pm \;=\; \mp\,u^2\,q_0^3 \sin(d/2).$$

In particular $q_+ - q_- = 2u^2 q_0 \sin(d/2)$, $q'_+ = q'_-$, and $q''_+ + q''_- = 0$. The phase difference $\Delta_m^{\mathrm{toy}}(s)$ at the symmetric pair is

$$\Delta_m^{\mathrm{toy}}(s) \;:=\; \Phi_{\mathrm{toy}}(t_-) - \Phi_{\mathrm{toy}}(t_+) \;=\; -q_0\,s \;=\; -d \tag{5.4}$$

(no u-dependence: the cosine is even at $\pm s/2$).

Proof. (5.3) substitutes $\tau = \pm s/2$ into (5.2) and uses oddness of sin and evenness of cos. For (5.4), evaluate (5.1) at $\tau = \pm s/2$:

$$\Phi_{\mathrm{toy}}(\pm s/2) = \pm\tfrac{q_0 s}{2} + u^2\big(1 - \cos(\mp d/2)\big) = \pm\tfrac{q_0 s}{2} + u^2\big(1 - \cos(d/2)\big),$$

and subtract. ☐

5.2. **Toy whitened block.**

Definition 5.5 (Toy finite blocks)**.** For $(m, s) \in \mathcal{D}_T$, with $t_\pm = m \pm s/2$, and for a frequency parameter $q_0 > 0$, evaluate the toy phase $\Phi_{\rm toy}(t; u, m_0, q_0)$ of Definition 5.1 with centre $m_0 = m$. Set

$$G_{{\rm toy},\pm}(s; m, u, q_0) \;:=\; G_{m,\pm}(s)\big|_{\Phi=\Phi_{\rm toy}}, \qquad N_{\rm toy}(s; m, u, q_0) \;:=\; N_m(s)\big|_{\Phi=\Phi_{\rm toy}}. \tag{5.5}$$

Here the right-hand sides are the same-point and mixed blocks (4.1)–(4.2) of Section 4 evaluated with (q, q', q'', Δ_m) replaced by $(q_{\rm toy}, q'_{\rm toy}, q''_{\rm toy}, \Delta_m^{\rm toy})$ of Lemmas 5.3–5.4.

Definition 5.6 (Toy whitened block)**.** For $(m, s) \in \mathcal{D}_T$, $q_0 > 0$, and $G_{{\rm toy},\pm}(s; m, u, q_0) \succ 0$, define

$$\widehat{\Omega}_{\rm toy}(s; m, u, q_0) \;:=\; G_{{\rm toy},-}(s; m, u, q_0)^{-1/2}\, N_{\rm toy}(s; m, u, q_0)\, G_{{\rm toy},+}(s; m, u, q_0)^{-1/2}. \tag{5.6}$$

The inverse square root is the unique positive definite one.

Lemma 5.7 (Toy block positivity)**.** *Fix a compact interval $D = [d_-, d_+] \subset (0, \infty)$. There exist $u_G = u_G(D) \in (0, \frac{1}{2}]$, $Q_G(D) \geq 1$, and $c_G(D) > 0$ such that whenever $d = q_0\, s \in D$, $|u| \leq u_G$, and $q_0 \geq Q_G(D)$,*

$$G_{{\rm toy},\pm}(s; m, u, q_0) \succ 0, \qquad \lambda_{\min}\big(G_{{\rm toy},\pm}(s; m, u, q_0)\big) \;\geq\; c_G(D)\, q_0.$$

In particular $\|G_{{\rm toy},\pm}(s; m, u, q_0)^{-1/2}\|_{\rm op} \leq C_G(D) q_0^{-1/2}$ on the same region.

Proof. Write $r = u^2$, $z = \sin(d/2)$, and $c = \cos(d/2)$. From (5.3), after factoring out the powers of q_0, the normalized same-point determinant has the form

$$q_0^{-4} D_J(t_\pm) = 4(1 \pm rz)^4 \mp 2rz(1 \pm rz) - 3r^2c^2.$$

This expression converges uniformly to 4 as $r \to 0$, uniformly in $d \in D$. Decrease u_G, if necessary, so that $q_0^{-4} D_J(t_\pm) \geq 1$ and $q_\pm \geq q_0/2$ for $|u| \leq u_G$. By Lemma 3.5, $G_{{\rm toy},\pm} \succ 0$. Also, on $d \in D$ and $|u| \leq u_G$, $\operatorname{Tr} G_{{\rm toy},\pm} \leq C(D) q_0^3$. Therefore

$$\lambda_{\min}(G_{{\rm toy},\pm}) \geq \frac{\det G_{{\rm toy},\pm}}{\operatorname{Tr} G_{{\rm toy},\pm}} \geq c_G(D) q_0.$$

After increasing $Q_G(D)$, this holds throughout the retained high-frequency region. The inverse-square-root bound follows by taking reciprocals and square roots. $\square$

5.3. **Toy anomaly functional and leading expansion.**

Definition 5.8 (Toy anomaly functional)**.** For a real 2×2 matrix $X = (x_{ij})$, the toy anomaly functional is the off-diagonal trace

$$\mathcal{A}_{\rm toy}(X) \;:=\; x_{12} + x_{21}. \tag{5.7}$$

Lemma 5.9 (Vanishing on the on-line baseline)**.** *At $u = 0$ the toy phase reduces to the linear baseline $\Phi_{\rm toy}(t; 0, m, q_0) = q_0(t - m)$, and $\widehat{\Omega}_{\rm toy}(s; m, 0, q_0)$ is the constant-phase whitened block of the linear baseline. In particular*

$$\mathcal{A}_{\rm toy}\big(\widehat{\Omega}_{\rm toy}(s; m, 0, q_0)\big) \;=\; 0 \qquad \big((m, s) \in \mathcal{D}_T^\times\big). \tag{5.8}$$

Proof. At $u = 0$, (5.2) gives $q_\pm = q_0$ and $q'_\pm = q''_\pm = 0$; by (4.1), $G_{{\rm toy},\pm}(s; m, 0) = (1/\pi)\operatorname{diag}(2q_0, q_0^3/6)$ is diagonal, and $G_{{\rm toy},-}^{-1/2} = G_{{\rm toy},+}^{-1/2}$ is the same diagonal matrix. By (4.2) with $q_+ = q_- = q_0$,

$$N_{\rm toy}(s; m, 0, q_0)_{12} + N_{\rm toy}(s; m, 0, q_0)_{21} = \frac{(q_+ - q_-)\cos \Delta_m^{\rm toy}(s)}{\pi s} = 0.$$

Whitening by the diagonal $G^{-1/2}$ preserves this off-diagonal cancellation: $\widehat{\Omega}_{12} + \widehat{\Omega}_{21} = (G^{-1/2})_{11}(G^{-1/2})_{22}(N_{12} + N_{21}) = 0$. $\square$

Lemma 5.10 (Leading u^2 expansion of the off-diagonal trace)**.** *Fix a compact interval $D = [d_-, d_+] \subset (0, \infty)$. There exist $Q_1 = Q_1(D) \geq 1$, $u_1 = u_1(D) > 0$, and a function $F_{\mathrm{toy}} : D \times [Q_1, \infty) \to \mathbb{R}$ such that for every $0 < |u| < u_1$, every $q_0 \geq Q_1$, and every $(m, s) \in \mathcal{D}_T^\times$ with $d = q_0\, s \in D$,*

$$\mathcal{A}_{\mathrm{toy}}\big(\widehat{\Omega}_{\mathrm{toy}}(s; m, u, q_0)\big) \;=\; u^2\, F_{\mathrm{toy}}(d, q_0) \;+\; u^4\, R_{\mathrm{toy}}(d, q_0, u), \tag{5.9}$$

with $|R_{\mathrm{toy}}(d, q_0, u)| \leq C_R(D)$ uniformly on the region $\{|u| \leq u_1,\, d \in D,\, q_0 \geq Q_1(D)\}$. The high-frequency limit

$$F_\infty(d) \;:=\; \lim_{q_0 \to \infty} F_{\mathrm{toy}}(d, q_0) \tag{5.10}$$

exists, is reached uniformly on compact d-intervals, and admits the explicit closed form

$$\begin{aligned} F_\infty(d) \;=\; \frac{\sqrt{3}}{8\, d^3\, \sin(d/2)} \Big(& 2\, d^2 \cos d + 5\, d^2 \cos 2d - 7\, d^2 \\ & + 6\, d \sin d - 15\, d \sin 2d - 12 \cos 2d + 12 \Big). \end{aligned} \tag{5.11}$$

F_∞ is interpreted by removable continuation at the apparent $\sin(d/2)$*-poles $d = 2\pi k$, $k \geq 1$. Its zero set in $(0, \infty)$ is discrete. On every compact subinterval $D \subset (0, \infty)$ avoiding the zeros of this continued function there exist $Q_0(D) \geq 1$ and $c_{\mathrm{toy}}(D) > 0$ with*

$$\big|F_{\mathrm{toy}}(d, q_0)\big| \;\geq\; c_{\mathrm{toy}}(D) \qquad (d \in D,\; q_0 \geq Q_0(D)). \tag{5.12}$$

Proof. Each entry of $G_{\mathrm{toy},\pm}(s; m, u, q_0)$ and $N_{\mathrm{toy}}(s; m, u, q_0)$ is a real-analytic function of $r := u^2$ by (5.2) and (4.1)–(4.2). To make the uniformity in q_0 explicit after substituting $s = d/q_0$, use the natural powers q_0, q_0^2, q_0^3 attached to the value/derivative coordinates:

$$G_{11} = q_0\, g_{11}, \quad G_{12} = q_0^2\, g_{12}, \quad G_{22} = q_0^3\, g_{22},$$

and similarly

$$N_{11} = q_0\, n_{11}, \quad N_{12} = q_0^2\, n_{12}, \quad N_{21} = q_0^2\, n_{21}, \quad N_{22} = q_0^3\, n_{22}.$$

The coefficients g_{ij} and n_{ij} are real-analytic functions of (r, d, η), $\eta = q_0^{-1}$, on $[0, u_1^2] \times D \times [0, \eta_1]$, once u_1 is small and $Q_1 = \eta_1^{-1}$ is large. By Lemma 5.7, the normalized determinants of the $G_{\mathrm{toy},\pm}$ stay bounded away from zero on this compact parameter set. The explicit 2×2 positive-definite inverse-square-root formula therefore gives

$$(G_{\mathrm{toy},\pm}^{-1/2})_{11} = q_0^{-1/2} h_{11}^\pm, \quad (G_{\mathrm{toy},\pm}^{-1/2})_{12} = q_0^{-3/2} h_{12}^\pm, \quad (G_{\mathrm{toy},\pm}^{-1/2})_{22} = q_0^{-3/2} h_{22}^\pm,$$

where the $h_{ij}^\pm$ are again real-analytic and uniformly bounded on the same compact set. Multiplying the three factors in $G_{\mathrm{toy},-}^{-1/2} N_{\mathrm{toy}} G_{\mathrm{toy},+}^{-1/2}$ shows that every entry of $\widehat{\Omega}_{\mathrm{toy}}$, and in particular $\mathcal{A}_{\mathrm{toy}}(\widehat{\Omega}_{\mathrm{toy}})$, is real-analytic in r with uniformly bounded Taylor coefficients. Since the u^0 coefficient vanishes by Lemma 5.9, the leading nonzero term is at order u^2; call its coefficient $F_{\mathrm{toy}}(d, q_0)$.

Direct computation using the closed-form 2×2 SPD inverse-square-root, applied to $G_{\mathrm{toy},\pm} = G_{\mathrm{toy},\pm}^{(0)} + u^2\, G_{\mathrm{toy},\pm}^{(2)} + O(u^4)$ of (4.1) with the toy substitutions of (5.3) and to N_{toy} of Definition 5.5, yields an explicit expression for $F_{\mathrm{toy}}(d, q_0)$ in $(\sin d, \cos d, \sin(d/2), \cos(d/2), q_0)$. After choosing $Q_1(D)$ large enough, this expression is regular for $d \in D$, $q_0 \geq Q_1(D)$, has a finite $q_0 \to \infty$ limit uniformly on D, and the limiting expression is (5.11).

The factor $\sin(d/2)$ of the denominator of (5.11) vanishes at $d = 2\pi k$, $k \in \mathbb{Z}_{\geq 1}$; the trigonometric numerator vanishes at the same points. Expanding the numerator at each $d = 2\pi k$ shows a factor of $\sin(d/2)$, so the apparent poles are removable and F_∞ extends real-analytically to $(0, \infty)$. On any compact subinterval avoiding the zeros of F_∞, the difference $F_{\mathrm{toy}}(d, q_0) - F_\infty(d)$ is $O(1/q_0)$ uniformly; for q_0 sufficiently large $|F_{\mathrm{toy}}(d, q_0)| \geq \frac{1}{2}|F_\infty(d)|$, giving (5.12). The remainder R_{toy} is the u^4-and-higher part of this real-analytic expansion. The compactness of the normalized parameter set and the uniform positive-definiteness just described give a Cauchy-type bound for these coefficients, uniformly for $d \in D$, $|u| \leq u_1$, and $q_0 \geq Q_1(D)$, after decreasing u_1 and increasing $Q_1(D)$ if necessary. □

5.4. **Toy retained subdomain and polynomial-weight density.**

Definition 5.11 (Toy retained subdomain)**.** Fix a compact interval $D = [d_-, d_+] \subset (0, \infty)$ avoiding the isolated zeros of F_∞ of Lemma 5.10. The signed toy retained subdomain at height T is

$$\mathcal{D}_T^{\mathrm{toy}}(D) := \big\{(m, s) \in \mathcal{D}_T^{\times} : q(m)\,|s| \in D\big\}. \tag{5.13}$$

Thus the positive and negative separation branches are both retained. The comparison estimates below are stated for the absolute scalar functional $|\mathcal{A}_{\mathrm{toy}}|$, so retaining both signs does not change the toy lower bound; see Lemma 5.13.

Lemma 5.12 (Polynomial-weight density of the toy retained subdomain)**.** *Let*

$$q_T^- := \inf_{m \in I_T} q(m), \qquad q_T^+ := \sup_{m \in I_T} q(m).$$

Assume $q_T^- > 0$ and

$$\frac{d_+}{q_T^-} \le \frac{|I_T|}{4}. \tag{5.14}$$

Then

$$\frac{|\mathcal{D}_T^{\mathrm{toy}}(D)|}{|\mathcal{D}_T|} \ge \frac{d_+ - d_-}{q_T^+\,|I_T|}. \tag{5.15}$$

Consequently, if the retained windows satisfy

$$|I_T| \le C_I Q(T)^{-1}, \qquad q_T^+ \le C_q q(T) \tag{5.16}$$

with absolute constants C_I, C_q, then

$$\frac{|\mathcal{D}_T^{\mathrm{toy}}(D)|}{|\mathcal{D}_T|} \ge \frac{d_+ - d_-}{C_I C_q}\,\frac{Q(T)}{q(T)}. \tag{5.17}$$

The height hypothesis (5.14) *is implied for all sufficiently large T in any scale regime with $|I_T| \ge c_I Q(T)^{-1}$, $q_T^- \ge c_q q(T)$, and $Q(T) = o(q(T))$, with constants depending only on D, c_I, c_q.*

Proof. The finite packet domain is

$$\mathcal{D}_T = \big\{(m, s) : |m - T| + |s|/2 \le |I_T|/2\big\},$$

a rotated square of area $|\mathcal{D}_T| = |I_T|^2$. Restrict to the central slab

$$S_T := \{m : |m - T| \le |I_T|/4\}.$$

For $m \in S_T$, the $\mathcal{D}_T$-constraint permits the full range

$$|s| \le |I_T| - 2|m - T|.$$

In particular it contains the smaller symmetric range $|s| \le |I_T|/4$. By (5.14),

$$\frac{d_+}{q(m)} \le \frac{d_+}{q_T^-} \le \frac{|I_T|}{4},$$

so both intervals

$$\left[\frac{d_-}{q(m)}, \frac{d_+}{q(m)}\right], \qquad -\left[\frac{d_+}{q(m)}, \frac{d_-}{q(m)}\right]$$

are contained in the allowed s-range. Their combined length is $2(d_+ - d_-)/q(m)$, which is at least $2(d_+ - d_-)/q_T^+$. Therefore

$$|\mathcal{D}_T^{\mathrm{toy}}(D)| \ge \int_{S_T} \frac{2(d_+ - d_-)}{q_T^+}\,dm = \frac{(d_+ - d_-)|I_T|}{q_T^+}.$$

Dividing by $|\mathcal{D}_T| = |I_T|^2$ proves (5.15). The estimate (5.17) follows immediately from (5.16). Finally, if $|I_T| \ge c_I Q(T)^{-1}$ and $q_T^- \ge c_q q(T)$, then (5.14) follows from

$$\frac{d_+}{c_q q(T)} \le \frac{c_I}{4Q(T)}, \quad \text{i.e.} \quad \frac{Q(T)}{q(T)} \le \frac{c_I c_q}{4d_+},$$

which holds eventually whenever $Q(T) = o(q(T))$. $\square$

Lemma 5.13 (Signed-separation invariance of the scalar toy channel)**.** *For the toy finite blocks of Definition 5.5, with fixed $q_0 > 0$,*

$$\widehat{\Omega}_{\mathrm{toy}}(-s; m, u, q_0) = \widehat{\Omega}_{\mathrm{toy}}(s; m, u, q_0)^{\top}$$

whenever both sides are defined. Consequently

$$\mathcal{A}_{\mathrm{toy}}(\widehat{\Omega}_{\mathrm{toy}}(-s; m, u, q_0)) = \mathcal{A}_{\mathrm{toy}}(\widehat{\Omega}_{\mathrm{toy}}(s; m, u, q_0)). \tag{5.18}$$

Proof. Changing s to $-s$ swaps the two centers t_- and t_+. The same-point blocks therefore satisfy

$$G_{\mathrm{toy},-}(-s; m, u, q_0) = G_{\mathrm{toy},+}(s; m, u, q_0), \qquad G_{\mathrm{toy},+}(-s; m, u, q_0) = G_{\mathrm{toy},-}(s; m, u, q_0),$$

and the mixed block satisfies $N_{\mathrm{toy}}(-s; m, u, q_0) = N_{\mathrm{toy}}(s; m, u, q_0)^{\top}$, by the symmetry of K_{Φ} and the row/column convention of Definition 4.2. Positive definite inverse square roots commute with transposition for symmetric matrices, so the displayed transpose identity for $\widehat{\Omega}_{\mathrm{toy}}$ follows. Since $\mathcal{A}_{\mathrm{toy}}(X) = X_{12} + X_{21}$ is invariant under transpose, (5.18) follows. $\square$

5.5. Toy anomaly visibility.

Theorem 5.14 (Toy anomaly visibility)**.** *Let $D \subset (0, \infty)$ be a compact interval avoiding the isolated zeros of F_{∞} of Lemma 5.10. There are constants $Q_{\mathrm{toy}}(D) \geq 1$ and $u_{\mathrm{toy}}(D) > 0$ such that, for every off-line displacement $u = \beta - \frac{1}{2} \in \mathbb{R}$ with $0 < |u| \leq u_{\mathrm{toy}}(D)$, every $q_0 \geq Q_{\mathrm{toy}}(D)$, and every $(m, s) \in \mathcal{D}_T^{\times}$ with $d = q_0\, s \in D$, the toy whitened block $\widehat{\Omega}_{\mathrm{toy}}(s; m, u, q_0)$ of Definition 5.6 satisfies*

$$\left|\mathcal{A}_{\mathrm{toy}}\big(\widehat{\Omega}_{\mathrm{toy}}(s; m, u, q_0)\big)\right| \;\geq\; \tfrac{1}{2}\, c_{\mathrm{toy}}(D)\, u^2. \tag{5.19}$$

The constant $c_{\mathrm{toy}}(D) > 0$ is the lower bound of Lemma 5.10 and depends on D only. On the polynomial-weight subset of midpoints and separations on which $d \in D$, the visibility exponent is $A_{\mathrm{toy}} = 0$: the lower bound is $(c_{\mathrm{toy}}(D)/2)\, u^2\, Q^0$, independent of the height.

Proof. Choose $Q_{\mathrm{toy}}(D) \geq \max\{Q_1(D), Q_G(D)\}$. Let $u_G(D)$ be the smallness threshold from Lemma 5.7. Choose

$$u_{\mathrm{toy}}(D) \leq \min\{u_1(D), u_G(D)\}$$

small enough that both lemmas apply. By (5.9) and (5.12),

$$\left|\mathcal{A}_{\mathrm{toy}}\big(\widehat{\Omega}_{\mathrm{toy}}(s; m, u, q_0)\big)\right| \;\geq\; u^2\big(c_{\mathrm{toy}}(D) - u^2\, C_R(D)\big).$$

Decreasing $u_{\mathrm{toy}}(D)$ so that $u_{\mathrm{toy}}(D)^2\, C_R(D) \leq c_{\mathrm{toy}}(D)/2$ gives (5.19). $\square$

Corollary 5.15 (Visibility on the polynomial-weight subdomain)**.** *Fix D and the constants $Q_{\mathrm{toy}}(D)$, $u_{\mathrm{toy}}(D)$, $c_{\mathrm{toy}}(D)$ of Theorem 5.14. Assume the height is large enough that $q_T^- \geq Q_{\mathrm{toy}}(D)$, where $q_T^- := \inf_{m \in I_T} q(m)$. On the toy retained subdomain we set $q_0 = q(m)$. Then for every $0 < |u| \leq u_{\mathrm{toy}}(D)$ and every $(m, s) \in \mathcal{D}_T^{\mathrm{toy}}(D)$,*

$$\left|\mathcal{A}_{\mathrm{toy}}(\widehat{\Omega}_{\mathrm{toy}}(s; m, u, q(m)))\right| \geq \tfrac{1}{2} c_{\mathrm{toy}}(D) u^2. \tag{5.20}$$

If, in addition, the hypotheses of Lemma 5.12 hold, then this lower bound holds on a subset of $\mathcal{D}_T$ of relative density at least the right-hand side of (5.15)*, and hence at least* (5.17) *under* (5.16)*.*

Proof. Let $(m, s) \in \mathcal{D}_T^{\mathrm{toy}}(D)$. Then $d = q(m)|s| \in D$ and $q_0 = q(m) \geq q_T^- \geq Q_{\mathrm{toy}}(D)$. If $s > 0$, Theorem 5.14 gives (5.20). If $s < 0$, apply the theorem to $|s|$ and then use Lemma 5.13. The density statement is Lemma 5.12. $\square$

Remark 5.16 (Transverse projection deferred)**.** The functional $\mathcal{A}_{\mathrm{toy}}$ annihilates the on-line baseline (Lemma 5.9). The later projection Π_{trans} must be chosen so that this scalar functional descends to or is represented on the retained transverse quotient. Once Section 6 proves that compatibility, the visibility bound (5.19) transfers to the Π_{trans}-projected comparison with the same exponent $A_{\mathrm{toy}} = 0$. This compatibility is not proved in the present section.

Remark 5.17 (Compatibility with the archived v1 toy matrix)**.** The toy phase (5.1) is a replacement toy for the clean scalar-functional rebuild. It is not the archived v1 matrix $M(d)$, and this section does not import the v1 calibration map $\Psi_{x,d}$ or the Frobenius pairings with A_{val}. Consequently this change is isolated for the present rebuild through Sections 6–8, where only the scalar functional $\mathcal{A}_{\text{toy}}$, the exponent $A_{\text{toy}} = 0$, and the lower bound (5.19) are consumed. Any later attempt to import a v1 theorem that explicitly uses $M(d)$, $\Phi_K(M(d))$, A_{val}, or $\Psi_{x,d}$ must be rederived for the present toy phase or kept in the archive.

Remark 5.18 (No uniform extension; polynomial-weight scope)**.** The visibility lower bound (5.19) cannot be extended to all of $\mathcal{D}_T$. At $s = 0$, the same coalescence argument as Corollary 4.7, applied to the toy finite blocks, gives $\widehat{\Omega}_{\text{toy}}(0; m, u, q_0) = I_2$ for every u. Hence $\mathcal{A}_{\text{toy}}(\widehat{\Omega}_{\text{toy}}(0; m, u, q_0)) = 0$; no u^2-lower bound can hold at the coalescence locus $d = 0$. The regimes $d \to \infty$ and d approaching an isolated zero of F_∞ are excluded by the compactness and avoidance hypotheses on D in Definition 5.11. The signed polynomial-weight subdomain $\mathcal{D}_T^{\text{toy}}(D)$ carries the toy visibility lower bound by Corollary 5.15; its relative density is polynomial in Q by Lemma 5.12 and is the only density loss contributed by this replacement of the impossible all-of-$\mathcal{D}_T$ assertion.

6. Transverse projection

paper: [**LIVE**] v1: [**migrated**] referee: [**in-progress**] sympy: [**verified**] `rh/sympy/sec-transverse-projection.py`
sim: [**verified**] `rh/simulation/sec-transverse-projection.py` lean: [**not-started**]

The scalar toy anomaly functional $\mathcal{A}_{\text{toy}}$ of Definition 5.8 fixes the two-point value subspace as its kernel. The transverse projection in this section is the Hilbert–Schmidt orthogonal projection of the Gram-normalized 2×2 matrix space onto the one-dimensional complement of that kernel. This gives a canonical two-point Π_{trans} compatible with the scalar visibility lower bound of Theorem 5.14. The projection has Hilbert–Schmidt operator norm one, contributes no polynomial loss, and allows the actual-zeta two-point suppression statement of Section 7 to be formulated in the same scalar channel as the toy lower bound. All occurrences of Π_{trans} in this section are therefore understood as two-point objects on $M_2(\mathbb{R})$. This section does not define the four-point projection $\Pi^{(4)}_{\text{trans}}$.

6.1. **Value subspace and transverse projection.** We work on $M_2(\mathbb{R})$ with the Hilbert–Schmidt inner product $\langle X, Y\rangle_{\text{HS}} := \operatorname{tr}(X^\top Y)$.

Definition 6.1 (Value subspace)**.** The two-point scalar value subspace is

$$\mathcal{V}_{\text{val}} := \{X \in M_2(\mathbb{R}) : x_{12} + x_{21} = 0\} = \ker \mathcal{A}_{\text{toy}}. \tag{6.1}$$

Lemma 6.2 (Explicit basis for $\mathcal{V}_{\text{val}}$)**.** *$\mathcal{V}_{\text{val}}$ is a three-dimensional subspace of $M_2(\mathbb{R})$ with orthonormal basis*

$$e_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad e_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad e_{\text{anti}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{6.2}$$

Its one-dimensional orthogonal complement is

$$\mathcal{V}_{\text{val}}^\perp = \mathbb{R}\, e_{\text{sym}}, \qquad e_{\text{sym}} := \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{6.3}$$

Proof. The four matrices $e_{11}, e_{22}, e_{\text{anti}}, e_{\text{sym}}$ form an orthonormal basis of $M_2(\mathbb{R})$ by direct calculation. The first three have $x_{12} + x_{21} = 0$, so they lie in $\mathcal{V}_{\text{val}}$. The matrix e_{sym} has $x_{12} + x_{21} = \sqrt{2} \neq 0$ and is orthogonal to the first three, so it spans $\mathcal{V}_{\text{val}}^\perp$. □

Remark 6.3 (Four-point projection not supplied here)**.** This section defines the two-center projection on $M_2(\mathbb{R})$. The four-center projection $\Pi^{(4)}_{\text{trans}}$ appearing in Hypothesis 7.3 and Theorem 8.6 is a separate object. No four-point quotient, four-point value subspace, or four-point scalar functional is proved in this section.

Definition 6.4 (Two-point transverse projection)**.** The two-point transverse projection $\Pi_{\text{trans}} : M_2(\mathbb{R}) \to M_2(\mathbb{R})$ is the Hilbert–Schmidt orthogonal projection onto $\mathcal{V}_{\text{val}}^{\perp}$. Equivalently,

$$\Pi_{\text{trans}} X \;=\; \frac{x_{12}+x_{21}}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}. \tag{6.4}$$

Definition 6.5 (Two-point scalar transverse channel)**.** For a real two-center whitened block $X \in M_2(\mathbb{R})$, define

$$\mathcal{A}_2(X) := \mathcal{A}_{\text{toy}}(\Pi_{\text{trans}} X). \tag{6.5}$$

Lemma 6.8 gives $\mathcal{A}_2(X) = \mathcal{A}_{\text{toy}}(X) = x_{12} + x_{21}$. Thus $\mathcal{A}_2$ is a scalar functional on two-point matrix representatives, or equivalently on the quotient $M_2(\mathbb{R})/\mathcal{V}_{\text{val}}$ after choosing the Hilbert–Schmidt orthogonal representative $\Pi_{\text{trans}} X$.

6.2. **Properties of Π_{trans}.**

Lemma 6.6 (Projection properties)**.** *The operator Π_{trans} is idempotent, self-adjoint with respect to the Hilbert–Schmidt inner product, has kernel exactly $\mathcal{V}_{\text{val}}$, and, as an operator on $(M_2(\mathbb{R}), \|\cdot\|_{\text{HS}})$,*

$$\|\Pi_{\text{trans}}\|_{\text{op,HS}\to\text{HS}} = 1. \tag{6.6}$$

Proof. Formula (6.4) is the same as $\Pi_{\text{trans}} X = \langle X, e_{\text{sym}}\rangle_{\text{HS}}\, e_{\text{sym}}$. Hence Π_{trans} is the Hilbert–Schmidt orthogonal projection onto the nonzero subspace $\mathbb{R}e_{\text{sym}}$. Idempotence, self-adjointness, kernel $\mathcal{V}_{\text{val}}$, and operator norm one follow immediately. ☐

Lemma 6.7 (Quotient representative and scalar norm)**.** *For every $X \in M_2(\mathbb{R})$,*

$$\Pi_{\text{trans}} X = \frac{\mathcal{A}_{\text{toy}}(X)}{\sqrt{2}}\, e_{\text{sym}}, \qquad X - \Pi_{\text{trans}} X \in \mathcal{V}_{\text{val}}, \tag{6.7}$$

and

$$\|\Pi_{\text{trans}} X\|_{\text{HS}}^2 = \frac{|\mathcal{A}_{\text{toy}}(X)|^2}{2}. \tag{6.8}$$

Proof. The representative identity follows from (6.4) and $e_{\text{sym}} = 2^{-1/2}\left(\begin{smallmatrix}0 & 1\\ 1 & 0\end{smallmatrix}\right)$. Applying $\mathcal{A}_{\text{toy}}$ to $X - \Pi_{\text{trans}} X$ gives zero by Lemma 6.8, so the residual lies in $\mathcal{V}_{\text{val}}$. The norm identity follows from $\|e_{\text{sym}}\|_{\text{HS}} = 1$. ☐

Lemma 6.8 (Scalar-channel compatibility)**.** *For every $X \in M_2(\mathbb{R})$,*

$$\mathcal{A}_{\text{toy}}(\Pi_{\text{trans}} X) \;=\; \mathcal{A}_{\text{toy}}(X). \tag{6.9}$$

Consequently the toy visibility lower bound (5.19) *is unchanged if the toy whitened block is first replaced by its projected representative $\Pi_{\text{trans}}\widehat{\Omega}_{\text{toy}}$. No matrix-norm lower bound is asserted; only the scalar channel $\mathcal{A}_2 = \mathcal{A}_{\text{toy}} \circ \Pi_{\text{trans}}$ is transported.*

Proof. By (6.4), $(\Pi_{\text{trans}} X)_{12} = (\Pi_{\text{trans}} X)_{21} = (x_{12} + x_{21})/2$. Therefore

$$\mathcal{A}_{\text{toy}}(\Pi_{\text{trans}} X) \;=\; \frac{x_{12}+x_{21}}{2} + \frac{x_{12}+x_{21}}{2} \;=\; x_{12} + x_{21} \;=\; \mathcal{A}_{\text{toy}}(X).$$

The final assertion follows by applying this equality to $X = \widehat{\Omega}_{\text{toy}}(s; m, u, q_0)$ inside the absolute value in (5.19). ☐

Lemma 6.9 (Two-point scalar quotient well-definedness)**.** *The scalar channel $\mathcal{A}_2$ descends to the quotient $M_2(\mathbb{R})/\mathcal{V}_{\text{val}}$. Equivalently, if $X - Y \in \mathcal{V}_{\text{val}}$, then*

$$\mathcal{A}_2(X) = \mathcal{A}_2(Y). \tag{6.10}$$

Moreover, for every $X \in M_2(\mathbb{R})$,

$$\mathcal{A}_2(X) = \mathcal{A}_{\text{toy}}(\Pi_{\text{trans}} X) = \mathcal{A}_{\text{toy}}(X) = \sqrt{2}\,\langle X, e_{\text{sym}}\rangle_{\text{HS}}. \tag{6.11}$$

Proof. If $X - Y \in \mathcal{V}_{\text{val}} = \ker \mathcal{A}_{\text{toy}}$, then $\mathcal{A}_{\text{toy}}(X) - \mathcal{A}_{\text{toy}}(Y) = 0$. By Definition 6.5 and Lemma 6.8, $\mathcal{A}_2 = \mathcal{A}_{\text{toy}}$ on matrix representatives; hence (6.10). The identities in (6.11) follow from Lemma 6.8 and $\langle X, e_{\text{sym}}\rangle_{\text{HS}} = (x_{12} + x_{21})/\sqrt{2}$. ☐

6.3. Scope of the chosen projection.

Remark 6.10 (No richer quotient is used in the scalar rebuild)**.** The downstream two-point comparison consumes only the scalar visibility $|\mathcal{A}_2(\widehat{\Omega}_{\mathrm{toy}})|$ and the matching actual-zeta scalar upper bound $|\mathcal{A}_2(\widehat{\Omega}_\zeta)|$. At this scalar level, the kernel of $\mathcal{A}_{\mathrm{toy}}$ is the complete value subspace. The orthogonal projection of Definition 6.4 gives the canonical Hilbert–Schmidt representative of the quotient map $M_2(\mathbb{R}) \to M_2(\mathbb{R})/\mathcal{V}_{\mathrm{val}}$, with no polynomial loss. Richer matrix-level quotient constructions from the archived local-projective-rigidity machinery are not imported here; cf. Remark 5.17.

Remark 6.11 (Residual gauge)**.** The retained chart of Definition 2.4 pins $\Phi_T = \theta$. Additive constants $\Phi \mapsto \Phi + c$ leave q, q', q'', hence $G_{m,\pm}$, N_m, and $\widehat{\Omega}_\zeta$, unchanged. Affine-jet shifts $\Phi \mapsto \Phi + at$ change q to $q+a$ and do not preserve the chart class. Thus there is no residual gauge action on the two-point matrix representatives beyond the trivial additive-constant invariance.

Lemma 6.12 (Sufficient two-point projection input)**.** *The transverse projection Π_{trans} and scalar channel $\mathcal{A}_2$ supply the two-point projection input required by Section 8: the toy visibility lower bound* (5.19) *transports through Π_{trans} by Lemma 6.8, and the Hilbert–Schmidt operator-norm bound* (6.6) *contributes no polynomial-in-Q loss. The actual-zeta two-point upper bound must be stated in the same scalar channel, namely as a bound for $|\mathcal{A}_2(\widehat{\Omega}_\zeta(s;m))|$.*

Proof. Combine Lemmas 6.6, 6.7, 6.8, and 6.9 with Theorem 5.14 and Corollary 5.15. □

7. Actual-zeta transverse suppression

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The actual zeta phase Φ defines two whitened blocks — the two-center block $\widehat{\Omega}_\zeta(s;m)$ and a four-center analogue $\widehat{\Omega}_\zeta^{(4)}$ for the symmetric quartet — on which the actual-zeta transverse anomaly is suppressed at a strictly larger polynomial exponent than the corresponding toy lower bound of Theorem 5.14. In the two-point case, the transverse scalar channel is the one fixed in Section 6:

$$\mathcal{A}_2(X) \;:=\; \mathcal{A}_{\mathrm{toy}}(\Pi_{\mathrm{trans}} X) \;=\; \mathcal{A}_{\mathrm{toy}}(X).$$

The equality is Lemma 6.8. Thus Π_{trans} acts on the block before the scalar functional is applied; the expression $\Pi_{\mathrm{trans}}\mathcal{A}_2(X)$ is not used.

The two-point and four-point comparisons of Section 8 consume different forms of this suppression, stated separately below.

7.1. Two-point suppression.

Hypothesis 7.1 (Two-point actual-zeta transverse suppression)**.** For the two-center packet associated with ρ and a symmetry partner, after the gauge-fixed normalization of Definition 2.4,

$$\bigl|\mathcal{A}_2\bigl(\widehat{\Omega}_\zeta(s;m)\bigr)\bigr| \;\le\; Q^{-B_{\zeta,2}} \tag{7.1}$$

for an absolute exponent $B_{\zeta,2} > A_{\mathrm{toy}}$ on the retained subfamily matched to the toy visibility theorem. Equivalently, by Lemma 6.8,

$$\bigl|\mathcal{A}_{\mathrm{toy}}\bigl(\Pi_{\mathrm{trans}}\widehat{\Omega}_\zeta(s;m)\bigr)\bigr| \;=\; \bigl|\mathcal{A}_{\mathrm{toy}}\bigl(\widehat{\Omega}_\zeta(s;m)\bigr)\bigr| \;\le\; Q^{-B_{\zeta,2}}.$$

Gap 7.2. The two-point case of the structural gap. Consumed by Theorem 8.3. Admissible proof routes: (i) a standalone route from the local jet-Gram geometry alone; (ii) an energy-supply route consuming the source bound of Section 12 via Lemma 12.5. v1 prose to mine: the odd holomorphic transverse channel and corrected finite-s holomorphic whitening sections.

7.2. **Four-point suppression.**

Hypothesis 7.3 (Four-point actual-zeta transverse suppression)**.** For the four-center packet associated with the symmetry quartet $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$, let $\Pi^{(4)}_{\mathrm{trans}}$ be a four-point transverse projection and let $\mathcal{A}_{4,\mathrm{trans}}$ be the corresponding four-point transverse functional, i.e. a scalar or finite-dimensional normed output after the four-point projection has been applied. Then

$$\left\|\mathcal{A}_{4,\mathrm{trans}}\big(\widehat{\Omega}^{(4)}_{\zeta}\big)\right\| \;\le\; Q^{-B_{\zeta,4}} \tag{7.2}$$

for an absolute exponent $B_{\zeta,4} > A^{(4)}_{\mathrm{toy}}$ on the retained four-point subfamily.

Gap 7.4. The four-point case of the structural gap. Consumed by Theorem 8.6. Same admissible proof routes as Hypothesis 7.1. The projection $\Pi^{(4)}_{\mathrm{trans}}$, the four-point value subspace, and the functional $\mathcal{A}_{4,\mathrm{trans}}$ are not supplied by Section 6; they are separate four-center inputs. v1 prose to mine: the local projective rigidity of the one-pair and multi-pair families section, four-point variant.

Remark 7.5 (Standalone and energy-supply routes)**.** Hypotheses 7.1 and 7.3 are stated independently of the technology of their proofs. A purely local proof would use only the jet-Gram and quotient geometry of Sections 3–6; an energy-supply route may use the source-energy module of Section 12. Forward references from Section 8 indicate which hypothesis is invoked, not which route proves it.

8. Local projective rigidity

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Theorem 5.14 provides the toy lower bound and Hypotheses 7.1 and 7.3 provide the matching actual-zeta upper bounds on the transverse anomaly under Π_{trans} of Definition 6.4. The present section assembles the comparison. On the two-point family attached to ρ and a single symmetry partner, and on the four-point family of the full symmetric quartet $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$, the exponent inequality $A_{\mathrm{toy}} < B_{\zeta}$ (resp. its four-point analogue $A^{(4)}_{\mathrm{toy}} < B^{(4)}_{\zeta}$) is incompatible with the existence of the configuration. The two cases share the comparison mechanism but differ in the underlying jet-Gram object: the two-point case uses the cross-block N_{12} of Lemma 3.2; the four-point case uses a four-center analogue $\widehat{\Omega}^{(4)}_{\zeta}$ and a four-point extension $\Pi^{(4)}_{\mathrm{trans}}$ of the projection of Definition 6.4. Either case suffices for the aggregation step in Section 10.

8.1. **Branch exhaustion.**

Hypothesis 8.1 (Branch exhaustion)**.** Every admissible off-line packet produced by the branch-entry construction is excluded by at least one of:

(i) the two-point scalar branch of Theorem 8.3; or
(ii) the four-point full-quartet branch of Theorem 8.6.

Remark 8.2 (Status of Hypothesis 8.1)**.** Hypothesis 8.1 is not a consequence of Definition 6.4 and is not implied by Sections 6–7. The master conditional theorem of Section 1.2 consumes Hypothesis 8.1 alongside Theorems 8.3 and 8.6.

8.2. **Two-point comparison.** The two-point comparison takes ρ and one symmetry partner (typically $\bar{\rho}$ or $1-\rho$) and reads off the joint contribution to $N_m(s)$ at matched midpoint and separation. The exponent inequality $A_{\mathrm{toy}} < B_{\zeta}$ excludes the pair.

Theorem 8.3 (Two-point rigidity)**.** *Let $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$ and $|\gamma|$ sufficiently large, and let (T_1, T_2) be the height pair attached to ρ and one of its symmetry partners. Under Lemmas 3.1, 3.2, 4.8, Theorem 5.14, and Hypothesis 7.1,*

$$A_{\mathrm{toy}} < B_{\zeta,2} \qquad \Longrightarrow \qquad \rho \notin \{\zeta = 0\}.$$

Gap 8.4. Compose Theorem 5.14 (lower bound) with Hypothesis 7.1 (upper bound) at matched (m, s); show the exponent gap survives the polynomial whitening loss (Lemma 4.8) and the polynomial Gram lower bound (Lemma 3.6). The proof is short — arithmetic on absolute exponents — once the inputs are in place.

Remark 8.5 (Energy-supply route, if invoked)**.** If Hypothesis 7.1 cannot be discharged from the local jet-Gram geometry alone for the two-point case, the source-energy lower bound Hypothesis 12.3 of Section 12, transferred via Lemma 12.5, supplies the missing polynomial-in-Q room. Theorem 8.3 is then conditional on that source as well as on the local inputs.

8.3. **Four-point comparison in the symmetric quartet.** The four-point comparison uses the full symmetry packet $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$. The four-center jet-Gram block carries strictly more rigidity than the two-point cross-block: the toy visibility lower bound holds in a refined quotient invisible to the two-point comparison, and the matching suppression hypothesis is correspondingly stronger. The four-point theorem is stated independently of Theorem 8.3.

Theorem 8.6 (Four-point rigidity)**.** *Let ρ be a hypothetical off-line zero and $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$ its symmetry quartet. Under the four-point analogue of Theorem 5.14 and Hypothesis 7.3,*

$$A_{\mathrm{toy}}^{(4)} < B_{\zeta,4} \qquad \Longrightarrow \qquad \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\} \not\subset \{\zeta = 0\}.$$

Gap 8.7. State and prove the four-center analogue $\widehat{\Omega}_\zeta^{(4)}$ of $\widehat{\Omega}_\zeta$; the four-point projection $\Pi_{\mathrm{trans}}^{(4)}$; and the four-point analogue of Theorem 5.14 matching Hypothesis 7.3. Assemble the exponent inequality. v1 prose to mine: the local projective rigidity of the one-pair and multi-pair families section, together with the corrected fixed-target quotient package.

Remark 8.8 (Energy-supply route, if invoked)**.** The four-point analogue of Remark 8.5 applies: if Hypothesis 7.3 cannot be proved from local geometry alone, Hypothesis 12.3 of Section 12 supplies the missing room via the four-point branch of Lemma 12.5, separating $A_{\mathrm{toy}}^{(4)}$ from $B_{\zeta,4}$.

9. N-point odd-moment cancellation

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The two-point and four-point comparisons of Sections 8.2 and 8.3 carry finite Taylor remainders of Φ at packet centers. An explicit discrete N-point measure $\mu_{N,Q}$ on s-space, with its low-order odd moments cancelled by construction, absorbs those remainders below $\min(Q^{-A_{\mathrm{toy}}}, Q^{-B_\zeta})$ uniformly in the height.

Lemma 9.1 (N-point odd-moment cancellation)**.** *Let $\widehat{\mu}_{N,Q}$ be the symmetric N-point packet measure attached to height T and scale Q. For every $r \geq 0$,*

$$\partial_z^{2r+1} \widehat{\mu}_{N,Q}(0) = 0. \tag{9.1}$$

The same identity holds with $\widehat{\mu}_{N,Q}$ replaced by its convolution with any even bump $\widehat{\phi}$: for every $r \geq 0$,

$$\partial_z^{2r+1} \big(\widehat{\phi}(\eta z)\widehat{\mu}_{N,Q}(z)\big)\big|_{z=0} = 0. \tag{9.2}$$

Gap 9.2. Pure parity argument; v1 carries the full proof in the N-point odd-moment cancellation section. Migration is straightforward provided the symmetry packet of (9.1) matches Definition 2.4.

10. Aggregation and zero-free strip

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The two-point and four-point rigidity theorems exclude any single off-line zero above height T_0; the upgrade to a zero-free strip proceeds by aggregating along the dyadic height ladder, exploiting the polynomial nature of every loss in Lemmas 4.8, 3.6, Theorem 5.14, and Hypotheses 7.1 and 7.3.

Theorem 10.1 (Zero-free strip)**.** *There is* $T_0 > 0$ *such that* $\zeta(\beta + i\gamma) \neq 0$ *for every* $\beta \neq \frac{1}{2}$ *and* $|\gamma| \geq T_0$.

Gap 10.2. Aggregate Theorem 8.3 or Theorem 8.6 across the dyadic ladder; the polynomial losses combine into a single polynomial-in-$Q(\gamma)$ factor, and the exponent inequality $A_{\text{toy}} < B_\zeta$ carries through uniformly.

11. The Riemann hypothesis

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The dyadic zero-free strip of Theorem 10.1 excludes any off-line zero above the absolute height cutoff T_0; at heights below T_0 the standard numerical verification of RH covers the remaining zeros, completing the unconditional statement.

Theorem 11.1 (Riemann hypothesis)**.** *Every nontrivial zero of* ζ *satisfies* $\operatorname{Re} s = \frac{1}{2}$.

Proof. By Theorem 10.1, no off-line zero can lie at height $|\gamma| \geq T_0$; the standard low-height numerical verification of RH (see, e.g., [3, 5]) covers the range $|\gamma| < T_0$. □

12. Source-energy supply (optional)

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This section is self-contained: its sole role is to supply a single source-energy lower bound that Theorems 8.3 and 8.6 may invoke when their standalone forms in Section 8 are insufficient. Nothing earlier in the paper depends on this section.

12.1. **Canonical source curvature.**

Definition 12.1 (Canonical source curvature)**.** The canonical source curvature of ζ on the chart of Definition 2.4 is

$$q''_{\text{can}} := \partial_t^3 Z(t), \tag{12.1}$$

where $Z(t)$ is the real critical-line scalar of ζ in the same chart.

Gap 12.2. Identification of q''_{can} with $\partial_t^3 Z$ is a convention; chart-regularity and zero-separation hypotheses of Definition 2.4 carry over verbatim. v1 prose to mine: the upstream Gate-1 arithmetic inputs section.

12.2. **Source-energy bound.**

Hypothesis 12.3 (Source-energy lower bound)**.** On retained packet intervals $I \subset I_T$, the canonical source curvature satisfies

$$\int_I |q''_{\text{can}}(m)|^2 \, dm \;\geq\; Q^{-A_E} \, |I| \tag{12.2}$$

on a positive polynomial-weight subfamily, for an absolute exponent A_E.

Gap 12.4. polynomial lower bound on $\partial_t^3 Z$ across the retained packet family. This is an analytic estimate on ζ and not on the local jet-Gram geometry; it is what makes Section 12 optional. v1 prose to mine: the source-energy mass-good packet block.

12.3. **Energy-to-suppression transfer.**

Lemma 12.5 (Source energy to transverse suppression)**.** *Hypothesis 12.3, after carrier decomposition and packet trace normalization, implies one of Hypothesis 7.1 (with exponent* $B_{\zeta,2} = B_{\zeta,2}(A_E) > A_{\text{toy}}$*) or Hypothesis 7.3 (with exponent* $B_{\zeta,4} = B_{\zeta,4}(A_E) > A_{\text{toy}}^{(4)}$*), according to which carrier transfer is established into the corresponding transverse channel of* $\widehat{\Omega}_\zeta(s; m)$ *or* $\widehat{\Omega}_\zeta^{(4)}$.

Gap 12.6. Carrier decomposition and packet trace step taking $\int |q''_{\mathrm{can}}|^2$ on the source side into a lower bound on $\Pi_{\mathrm{trans}}\widehat{\Omega}_\zeta$ (two-point case) or $\Pi^{(4)}_{\mathrm{trans}}\widehat{\Omega}^{(4)}_\zeta$ (four-point case). v1 prose to mine: the certified downstream status after the canonical source-graded closure section, and the carrier-corrected source bridge package.

Appendix A. Archive map

paper: [**DIAGNOSTIC**] v1: [n/a] referee: [n/a] sympy: [n/a] sim: [n/a] lean: [n/a]

The previous large draft is preserved in place at `rh/proof_of_rh.tex` and is treated as a quarry from which prose and computations may be migrated, after restatement and refereeing, into the present file. Pointers to relevant v1 sections are given inline at each `wip` mark above; the table below summarizes the principal mining correspondences.

Rebuild section	**v1 source (in `proof_of_rh.tex`)**
§2	Local kernel and jet normalization.
§3	Finite-s local blocks; jet-limit local blocks.
§4	Corrected finite-s holomorphic whitening.
§5	Toy anomaly and transverse obstruction.
§6	Leading value matrix and canonical calibration.
§7	Odd holomorphic transverse channel; corrected fixed-target quotient package.
§8	Local projective rigidity of the one-pair and multi-pair families; finite coefficient table contracts.
§9	N-point odd-moment cancellation.
§10, §11	Master conditional theorem; introduction.
§12	Source-energy mass-good packet block; carrier-corrected source bridge package; upstream Gate-1 arithmetic inputs.

The v1 sections "Endgame architecture and remaining propagation problem outside the clean mixed four-point case," "Downstream local closure package: Gate 2, wall escape, and carrier rows," "Unified finite-survival architecture," "Current proof state after the local package audits," "How the older framing maps to the new framing," and "No-go and audit ledger" are deliberately not migrated; their content is route history and process bookkeeping rather than active proof material.

References

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